Date: . . . . No: . . . .

Lê Hoàng

N25DCCE051

Problem 1: 1,25
Problem 2: 2,5
Problem 3: 2,5
Problem 4: 2,0
8,25

P1: Singly linked List – In Place Reversal & Pointer Tracing

Task 1: Write a function to reverse the singly linked list to complete your function.

Task 2: Trace a diagram ar step by after each iteration until the list is fully

P2: Circular linked list – Insert at Head

Task 1: Write pseudocode for the function Insert Head to a node containing new value at the front of the circular

Task 2: Trace the execution steps when inserting new Value = 5 into the given list

P3: Doubly linked List – Insert Before Target & Edge Cases

Task 1 Write psudocode for Insert Before to insert new node

Task 2 list the exfact order of pointer specific

Task 3 Explicity handle the ~~foly~~ following

1 target is located at the head node

2 target is not ~~in the first~~ in the list

P4 Sorted linked list – In Oder Insertion

Given a singly list sorted [2] → [5] → [8] → [null]

Task 1: Write the function Sorted Insert to insert ascending order

Task 2: Construct a step-by-step trace table detailing pointer

Problem 1 : 1,25

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Task 1 :

0,0

```
Function  Insert Before (head, target, new value ) :
    current ← head
    While  current ≠ Null  & current.data ≠ target
        current ← current.next
    IF current = null
        Return head
    new Node ← creat new Node
    new Node . data ← new value

      ~~Rev~~ Node         current
    new ~~Node~~ . prev ← ~~current~~ . prev
    new . Node . next ← Current

    If current . prev ≠ Null
        current . prev . next ← new Node
    Else
        head ← new Node
    current . prev ← new Node
    Return Head
```

1,25 Task 2 : Pointer Trace

prev = null , curr = 10

B, S1 :

prev = 10 , curr = 25 , next = 25

10 → Null

S2

prev = 25 , curr = 40 , next = 40

25 → 10 → Null

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S3 : prev = 40, curr = 15, next = 15
~~25~~ → 40 → 25 → 10 → Null

S4 : prev = 15, curr = Null, next = Null
15 → 40 → 25 → 10 → Null

Propblem 2; 2,5

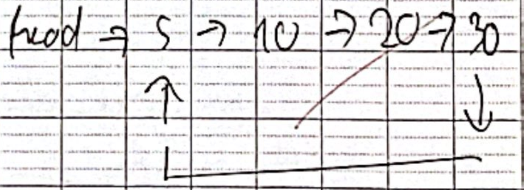
1,75 Task 1 : pseudocode

```
Insert (Head, new value)
  new Node ← create node (new value)
  If head = null
      new Node . next ← New node
      Return New Node
  tail ← head
  While (tail . next ≠ head)
        tail ← tail. next
  New node . next ← head
  tail. next ← new Node
  head ← New Node
Return head
```

0,75 Task 2

Initial
10 → 20 → 30 → 10
find tail :
tail = 30
new Node . next = head
5 → 10
tail . next = new Node
30 → 5
head = new Node

~~Result~~
Result
head → 5 → 10 → 20 → 30



Problem 3 : 2,5

0,75 Task 1 Pseudocode

```
Ins Before (head, target, new Value)
    curr ← head
    While curr ≠ null & curr . data ≠ target
        curr ← curr . next
    If curr = null
        Return head
    New node ← creat node (new Value)
    New node . prev ← curr . prev
    New node . next ← curr

    If (curr . prev ≠ null)
        curr . prev . next ← new Node
    ~~Else~~ Else :
        head ← new Node
        curr . prev ← new Node
    Return head
```

0,75 Task 2: Pointer assignment

```
New node . prev ← curr . prev
New node . next ← curr
curr . prev . next ← new Node
curr . prev ← New Node
```

1 Task 3 Edge cases

~~two~~ Case 1 Target = head

20 ⇄ 30 ⇄ 40

insert 10 before 20

10 ⇄ 20 ⇄ 30 ⇄ 40

Case 2 : Target = null

```
If curr = Null
    Return head
```

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Vũ Hoàng

N2SDECE0591

2,0 Problem 4:

1,00 Task 1: pseudocode

```
Sorted Insert (head, val)
New Node ← creat node (val)
If (head = null or val < head . data)
        new Node . next ← head
        head ← new Node
        Return head
    curr ← head
   While (curr . next ≠ null
        and curr . next . data < val)
        curr ← curr . next
   new Node . next ← curr . next
   curr . next ← new Node
          return head
```

1,0 Task 2 Trace

Initial

2 → 5 → 8 → Null

curr = 2

5 < 6 → True

→ curr = 5

8 < 6 → false

→ Stop

insert 6 before 5

New node . next = 8

curr next = 6

Result

head → [2] → [5] → [6]

→ [8] → Null